

Scheme of Studies for BS (Computer Science) | 2023

Deadlocks (Deadlock Characterization, Methods for Handling Deadlocks, Deadlock Prevention, Avoidance and Detection) Memory Management (Logical vs. Physical Address Space, Swapping, Contiguous Allocation, Paging, Segmentation, Segmentation with Paging) Virtual Memory (Demand Paging, Page Replacement, Page Replacement Algorithms, Allocation of frames, Thrashing) File System Interface and Implementation (File Concept, Access Methods, Directory Structure, Protection, File System Structure, Allocation Methods, Free Space Management, Directory Implementation)

Course Weekly Schedule

The course schedule for 17 weeks is detailed below

Week	Topic
1	Types of software, Application Software and System software, Application software categories, System software categories.
2	Introduction to OS, Single user, Multi user and Network OS, H/W dependent and independent OS, Important OS components, Kernel, Shell, Shell types
3	Type of services provided by OS, Uni- programming and Multi-programming, Uni-tasking and Multi-tasking, Uni- processing and Multi-processing.
4	Distributed systems and its types, Reason for building distributed systems,
5	SPOOLing, Buffering, Real-Time systems, Booting mechanism.
6	Process and process states. Process Control Block, Independent and Cooperating processes, Process and Threads, Hierarchy of processes.
7	Inter Process Communication, Process Synchronization, Race condition, Mutual exclusion, Critical section, Achieving process synchronization.
8	Scheduler and scheduling queues, Types of schedulers, CPU and I/O bound processes, Context switching, Criteria for comparing CPU scheduling algorithms.
9	Mid Term Exams
10	First Come First Served scheduling, Shortest Job First scheduling, Priority scheduling (Preemptive & Non-Preemptive), Round Robin scheduling.
11	Multi-level queue scheduling, Multi-level feedback queue (MLFBQ) scheduler, Multi-processor scheduling, and CPU Scheduling Algorithm evaluation.
12	Introduction to Deadlocks, Deadlock characterization, Deadlock prevention, Deadlock avoidance.
13	Introduction to Banker's Algorithm and importance of its usage, Deadlock detection in single resource of each resource type.
14	Deadlock detection in multiple resources of each resource type, Deadlock Recovery, Techniques used to recover processes from deadlock.